

Neuro-Symbolic Agents Playing Magic: The Gathering

Ontology-Grounded Knowledge Graphs, Latent World Models,
and Control-Theoretic Action Selection under Partial Observability

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Abstract

Magic: The Gathering (MTG) is, by several measures, the most combinatorially complex commercial card game ever published: more than 28,000 unique cards, a 260-page rule book, partial observability over 50–90 hidden cards, an unbounded action space, a turn-based stack with priority, and a state space conservatively estimated above 10^{800} . We study agents that play it *neuro-symbolically*: a verified rules engine decides what is legal, a formal ontology fixes what card semantics mean, a knowledge graph accumulates strategic structure, and learned components supply the statistics that no amount of symbolic knowledge can provide. Three claims organise the work. First, the symbolic layer should be a real ontology rather than a schema: we develop a provenance-aware OWL ontology of MTG aligned to DOLCE, with OWL 2 DL as its modelling target and its actual qualification boundaries described separately (Neubürger and Neubürger, 2026). Second, the knowledge graph should be *built by playing*: agents mine their own trajectories into an append-only, provenance-bearing extension layer that later agents query as a strategic prior, while curated card semantics remain untouched — a population-level learning loop in which the graph is shared long-term memory rather than static context. Third, action selection should be posed as *stochastic optimal control under partial observability* rather than as expected-free-energy minimisation: following Kaufmann (2026), the canonical discretisation of Active Inference is belief-space KL control, so we implement that objective directly and retain the two conventional expected-free-energy factorisations only as ablation arms with documented failure modes. We describe the architecture, state where the symbolic/sub-symbolic boundary lies and what crosses it in each direction, and specify the evaluation that would test the control claim inside MTG rather than on toy POMDPs. Code, ontology, and traces are released under the MIT licence.

Keywords: neuro-symbolic AI, knowledge graphs, world models, JEPA, stochastic optimal control, partially observable Markov decision processes, LLM agents, self-play, Magic: The Gathering.

1 Introduction

Reinforcement-learning (RL) research on games has historically progressed through a sequence of increasingly hard environments: backgammon (Tesauro, 1995), chess and Go (Silver et al., 2017, 2018), StarCraft II (Vinyals et al., 2019), and no-limit Texas hold’em poker (Brown and Sandholm, 2018). Each step pushed the community to invent new tools — temporal-difference (TD) learning, deep value networks, self-play, counterfactual regret minimisation (CFR). *Magic: The Gathering* (MTG) sits beyond the union of these challenges (Table 1): its action space is unbounded, its rules are open-textured natural language printed on cards (the comprehensive

Table 1: Approximate combinatorial complexity comparison.

	Chess	Go	No-limit Poker (HU)	MTG
State space	10^{47}	10^{170}	10^{160}	$> 10^{800}$
Branching factor	~ 35	~ 250	~ 10	0–100+ (unbounded)
Hidden information	none	none	2 cards	50–90+ cards
Card / piece pool	6 types	1 type	52 cards	$> 28,000$ unique
Rule text length	$\sim 10\text{p}$	$\sim 5\text{p}$	$\sim 10\text{p}$	$\sim 260\text{p}$ + per-card
Rules update frequency	decades	decades	decades	quarterly

rule book itself defers to the per-card text in many places), its hidden information includes the entire opponent’s deck composition, and its mechanics are extended every quarter by the publisher Wizards of the Coast. A single MTG card can override the rules of the game (e.g. *Yixlid Jailer*, *Blood Moon*, *Cumulative Upkeep*), and a single decision in combat may resolve dozens of triggered abilities in a player-chosen order. The state space is open in the strict sense: every quarterly set adds previously-unseen card text whose semantics must be parsed at runtime, so any agent that hard-codes its action vocabulary cannot survive a set rotation.

For a concrete sense of why this matters for an algorithmic agent: in a single midgame turn the active player may need to (i) decide whether to play a land, (ii) sequence two or three spells with mana cost, colour requirements and stack ordering, (iii) optionally activate any number of permanent abilities that themselves go on the stack, (iv) declare attackers under summoning-sickness and tapping rules, (v) respond to triggered abilities resolving in the order chosen by their controllers, and (vi) decide which damage to assign in combat under multi-blocker rules. Each of these is itself a sub-decision tree. The total number of legal actions per priority pass routinely exceeds 60 and, in pathological combo states, several hundred. A model-free policy that emits a fixed-size logit vector cannot represent this branching, and a tabular search procedure that enumerates the subtree cannot fit it into memory.

A purely model-free deep-RL approach is therefore unattractive: the action space cannot be enumerated as a fixed-size logit vector, the observation space is symbolic and irregular, and every new set introduces previously-unseen card text whose semantics must be parsed at run time. Conversely, a purely symbolic LLM-prompting approach struggles with rules-correctness, hallucinated cards, and the long-horizon credit-assignment that competitive play requires — an LLM that simply names a move it would like to make has no guarantee that move is legal, that it resolves the way the LLM expects, or that it furthers a coherent multi-turn plan.

Our position is that the structure of MTG demands a *neuro-symbolic* stack: a verified rules engine to keep play legal, a knowledge graph to make card semantics queryable, a learned world model to enable imagination and planning over the resulting compressed state, and an LLM agent that orchestrates these components under an active-inference objective. Figure 1 sketches the resulting architecture as a data-flow diagram, separating the *symbolic* substrate (data sources, engine, knowledge graph) from the *learned* substrate (world model, agent zoo) and showing where information actually crosses the boundary.

Contributions. This work makes four contributions.

1. A scientific framing of MTG as an *open-ended partially observable control problem* whose difficulty comes not only from combinatorics, but from the continual introduction of new symbolic rules through quarterly card releases.
2. An architecture in which the symbolic/sub-symbolic boundary is stated rather than assumed

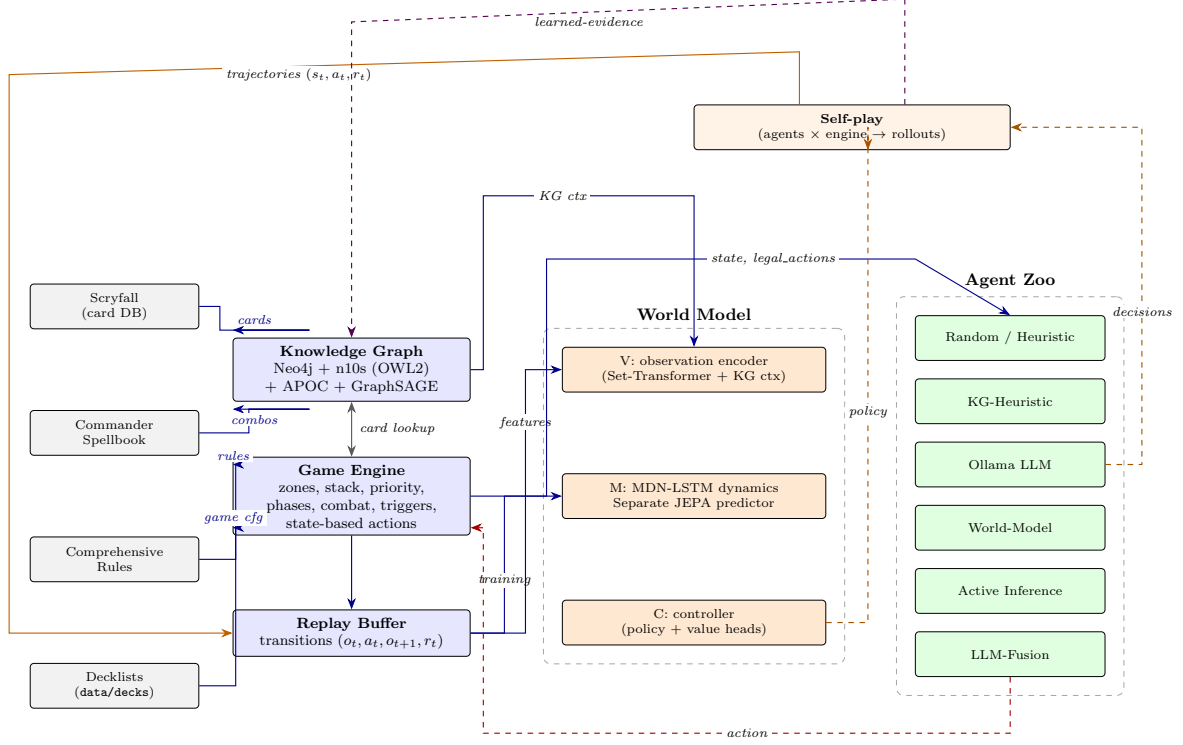


Figure 1: Data-flow architecture. **Grey** boxes are external data sources; **blue** the symbolic substrate (engine, knowledge graph, replay buffer); **orange** the learned world model (V+M+C with a JEPA target head and a knowledge-graph context channel); **green** the agent zoo. Solid blue arrows are forward data flow; the red dashed arrow is the agent’s chosen action returned to the engine; the engine $\leftrightarrow$ knowledge-graph link is bidirectional (card lookup at decision time, ontology-grounded card metadata at engine boot). Each agent additionally has read-only access to the knowledge graph and to the world-model controller through a thin lookup interface (omitted from the figure for clarity, described in the body text). The engine is the only component allowed to mutate game state; agents read but do not write.

(Section 3.5): what is symbolic, what is learned, what crosses in each direction, and what gates the crossing.

3. A *play-to-graph* learning loop in which agents mine their own trajectories into an append-only extension layer of the knowledge graph, and later agents query that layer as a strategic prior — with curated card semantics protected from statistical contamination by construction (Section 4.3).
4. A *control-theoretic* reading of MTG decision making: we adopt the result of Kaufmann (2026) that discrete Active Inference is belief-space KL control, implement that objective directly, and retain the two legacy expected-free-energy factorisations only as ablation arms (Section 3.3).

The ontology that grounds the symbolic layer is a contribution in its own right and is described in a companion paper (Neubürger and Neubürger, 2026); here we treat it as given and focus on what an agent does with it.

Why these four together. Each contribution targets a specific failure mode of the others. The engine prevents the LLM from playing illegal moves and the world model from imagining inconsistent dynamics. The knowledge graph compresses the 28,000-card pool into a structure with a few hundred archetype clusters so that an agent can reason about “what card type does

this state want next” without enumerating all candidates. The world model amortises the cost of multi-turn lookahead so that decisions remain well within a few hundred milliseconds even when planning ten turns deep. The active-inference fusion agent finally ties the three together with a single tunable objective — expected free energy — whose pragmatic and epistemic terms have a direct mapping onto MTG concepts of *value* and *information*.

Collective intelligence hypothesis. Beyond single-agent optimisation, the same stack naturally supports a population-level learning loop: many heterogeneous agents generate trajectories, those trajectories are mined for repeated card-pair and outcome regularities, and the resulting evidence is written into an append-only graph extension layer with provenance. New agents then query this enriched graph and inherit strategic priors discovered by previous runs. In this reading, the KG is not only static symbolic context but a shared long-term memory for the entire agent population.

Scope and what we do *not* do. We do not solve MTG; we do not even target tournament-strength play. The contribution is a research substrate that is rules-correct on the implemented card pool, that maintains a queryable symbolic representation of card semantics, and that supports plug-in research on world models and inference-style agents on a problem qualitatively harder than the standard benchmarks. The empirical claims in §6 are therefore framework-level (decks complete, agents converge faster than baselines, ablations show graded contributions) rather than absolute strength claims against human play.

The framework is released as open source¹ and the codebase is the canonical artifact accompanying this report. Section 2 positions the work against prior literature; §3 formalises the POMDP and the four learning principles (world models, FEP, knowledge-graph priors, constrained-decoding LLMs); §4 walks the architecture top-down; §6 presents the empirical protocol and preliminary results; §7–8 discuss limitations and next steps.

2 Related Work

World models. Ha and Schmidhuber (2018) introduced the V+M+C decomposition — a vision-VAE, recurrent mixture-density dynamics and a controller trained in imagined trajectories. Successors include Dreamer (Hafner et al., 2020, 2023) and the Schmidhuber-line of self-referential systems (Schmidhuber, 2015; Irie et al., 2022). We retain this separation but encode structured game observations with a Set-Transformer and optional symbolic context. A separate JEPA head (LeCun, 2022; Assran et al., 2023) learns action-conditioned next-embedding prediction, avoiding the requirement to generate a complete next-state description. LeWorldModel and the modular `stable-pretraining/stable-worldmodel` ecosystem provide related methods and reusable research infrastructure (Balestriero et al., 2025; Maes et al., 2026a,b). Representation prediction is a training principle, not itself a policy or a proof of strategically sufficient abstraction.

Active inference and stochastic optimal control. The Free-Energy Principle of Friston (2010) and its discrete-state operationalisation (Sajid et al., 2021) propose that action selection minimise *expected free energy* (EFE), conventionally decomposed into a pragmatic (reward-seeking) and an epistemic (information-seeking) term. Kaufmann (2026) shows that this framing inverts the actual mathematical hierarchy: continuous-time Active Inference on path space *is* finite-horizon path-integral stochastic MPC in the sense of Kappen (2005), Todorov (2009) and Williams et al. (2017), and its canonical discretisation is classical KL control on belief space. The exploration–exploitation trade-off is not supplied by an added epistemic term; it is Fel’dbaum’s dual control effect (Fel’dbaum, 1960), present in stochastic control from the outset, and it is

¹<https://github.com/cuteredpwnda/opposition-agents-playing-mtg>

recovered as soon as plans are evaluated *closed-loop* over beliefs rather than as open-loop action sequences. We follow that result rather than the conventional decomposition, for the concrete engineering reasons set out in Section 3.3: the two extant EFE factorisations have opposing, empirically demonstrated failure modes, and both are avoidable.

Knowledge graphs, GraphRAG, and their limits. Neo4j with the `n10s` plugin enables direct ingestion of OWL ontologies as labeled property graphs (Neo4j Labs, n.d.); the APOC library adds graph algorithms; GraphSAGE (Hamilton et al., 2017) provides inductive node embeddings. GraphRAG (Edge et al., 2024) complements vector RAG for queries that benefit from explicit relational structure, and Liao et al. (2026) give the first component-level ablation of a GraphRAG pipeline, finding that retriever choice and graph *serialisation format* dominate the outcome — structured, code-like serialisations outperform prose renderings of the same subgraph. MTG is a textbook GraphRAG case: “find me a two-card combo using only cards in my library that survive *Counterspell*” is a multi-hop graph query, not a similarity search. Two cautions from the same literature shape our design. Diettrich et al. (2026) show that consumers of an evolving schema must discover it at runtime rather than assume it, because static pipelines fail silently under schema drift; and Romanova (2026) gives a per-relation diagnostic for whether explicit graph structure still adds signal beyond node text at all, which is the honest way to decide where graph engineering is worth the effort.

Foundational ontologies and the symbolic layer. The intended symbolic substrate is a provenance-aware OWL ontology of MTG aligned to DOLCE-UltraLite (Gangemi et al., 2002; Borgo et al., 2022). Its conceptual schema and tooling were developed with LLM assistance; its normative vocabulary is mechanically derived from the Comprehensive Rules without runtime LLM inference. A companion paper (Neubürger and Neubürger, 2026) separates construction responsibilities and reports logical and data-validation scopes. Constraints are expressed as testable axioms, but schema consistency is not full profile conformance of the imported documents or certification of a populated agent graph. The broader turn toward competency-question-driven, validator-gated ontology construction (jiayi et al., 2026; Ontology Engineering Group, Universidad Politécnica de Madrid, 2026; Lippolis et al., 2026; Tiwari et al., 2026) is methodological context, not evidence that those external pipelines were executed here.

Neuro-symbolic systems and the curated/induced boundary. The framing that organises Section 3.5 is the distinction between *curated* and *induced* semantics. Sabou (2026) argues that trustworthy neuro-symbolic AI requires explanations grounded in domain knowledge rather than saliency, and identifies provenance and continuous auditing as the missing infrastructure. Thombre et al. (2026) demonstrate that neuro-symbolic decomposition — symbolic constraints narrowing the search space, the neural component reasoning inside each narrowed sub-task — substantially outperforms monolithic prompting. Van Schependom et al. (2026) generate reasoning training data from an ontology by backward-chaining proof search, the natural route to a reasoning benchmark *for* a schema such as ours. Ehrenmüller et al. (2026) address the reconciliation of conflicting graph claims from heterogeneous sources with explicit methodological provenance, which is structurally the same problem as reconciling curated card semantics with statistics mined from imperfect self-play. Our contribution to this thread is a domain where the induced layer is generated by an agent acting in a *verified* environment, so a learned assertion can in principle be traced back to the exact game states that support it.

MTG and other complex card games. Earlier MTG-AI work has used hand-crafted heuristics (Cowling et al., 2012), MCTS, and small policy nets on restricted card pools (e.g., `open-mtg`). On the rules side, `Forge`, `XMage` and `Mage` provide JVM-based reference implementations, and `phase-rs` provides a Rust one. To our knowledge, no prior public system combines a full

Comprehensive-Rules engine with a learned world model and a foundational-ontology-grounded symbolic layer. [Moravčík et al. \(2017\)](#) (poker), [Vinyals et al. \(2019\)](#) (StarCraft) and [Schrittwieser et al. \(2020\)](#) (MuZero) are spiritual antecedents at the algorithmic level.

Ontologies for games. Game ontologies support purposes including educational metadata, game design and simulation. The companion resource ([Neubürger and Neubürger, 2026](#)) targets interoperability between normative vocabulary, card descriptions, contextual gameplay and strategic evidence. Relative coverage and modelling commitments of other MTG representations have not yet been systematically compared; no priority claim is made.

3 Theoretical Foundations

3.1 MTG as a partially-observable Markov game

We model a game of Magic from the perspective of a single seat as a finite-horizon POMDP $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, \Omega, R, \gamma, b_0)$ embedded inside an n -player Markov game. The full state $s_t \in \mathcal{S}$ encompasses every public and private zone of every player — libraries, hands, the stack, mana pools, and a long tail of bookkeeping flags (summoning sickness, monarchy, the day/night state, designations, emblems, dungeon room, companion eligibility). A controlled-perspective filter $\Omega : \mathcal{S} \rightarrow \mathcal{O}$ hides the opposing players’ hands and the order of their libraries; the agent only ever observes $o_t = \Omega(s_t)$. The action space $\mathcal{A}_t = \text{LEGAL}(s_t)$ is computed afresh on every priority pass and is highly variable: typically $|\mathcal{A}_t| \in [1, 60]$, but can spike past several hundred in hard combo states. The transition model $T(s' | s, a)$ is the rules engine itself, fully deterministic given a random seed and the active player’s choices. Reward R is sparse and terminal — ± 1 on the win condition — but agents may shape it during training using Δlife , Δcards , Δtempo as auxiliary signals. The state space is combinatorially large: a conservative count over deck-times-zone configurations gives $|\mathcal{S}| > 10^{800}$, several orders of magnitude beyond Go.

The optimal value function over beliefs $b_t \in \Delta(\mathcal{S})$ satisfies the Bellman equation

$$V^*(b) = \max_{a \in \mathcal{A}} \left[\sum_s b(s) R(s, a) + \gamma \sum_o \Pr(o | b, a) V^*(b'_o) \right], \quad (1)$$

where b'_o is the Bayes update of b after taking a and observing o . Exact solution is intractable. Each agent in our zoo is therefore an explicit approximation that trades different axes of accuracy: tabular reactive policies (Random, Heuristic, KG-Heuristic, LLM) collapse b_t to a point estimate $s_t \approx o_t$; the world-model agent learns an amortised Bellman residual in latent space; the active-inference agents maintain an explicit belief and minimise expected free energy.

3.2 World models as latent imagination

The V+M+C decomposition of [Ha and Schmidhuber \(2018\)](#) separates representation learning, dynamics prediction and action selection. Our inputs are structured, perspective-filtered game observations: visible card identities and characteristics, zones, resources, turn information and public actions. The relevant choice is *predicting a compact representation of future observations versus predicting a complete next game-state description*.

What is predicted. An encoder E_ϕ maps an observation o_t and optional symbolic context g_t into a latent vector z_t . An action-conditioned predictor f_θ predicts the representation that the encoder would assign to the next observed position:

$$z_t = E_\phi(o_t, g_t), \quad \hat{z}_{t+1} = f_\theta(z_t, a_t), \quad z_{t+1}^{\text{target}} = \text{sg}(E_\phi(o_{t+1}, g_{t+1})). \quad (2)$$

Here ϕ and θ denote learned parameters, a_t is an encoded action, and sg means *stop gradient*: the target is treated as fixed for this loss evaluation. The current and target branches use the same encoder parameters; a separately maintained target network is not assumed. Training asks whether the consequence of an action is predictable in the learned representation, rather than demanding an exact listing of every future card, flag and hidden zone.

This is the joint-embedding predictive principle (LeCun, 2022; Assran et al., 2023; Maes et al., 2026b). It can allocate capacity to regularities useful for evaluating positions without decoding a full symbolic successor state. That is an inductive bias, not a guarantee: compression may discard a decisive interaction, and unpredictable hidden information does not become observable merely because a latent exists. The learned representation z_t is also distinct from a Bayesian belief b_t over possible hidden states.

What the training loss compares. The present JEPA objective uses consecutive transitions (o_t, a_t, o_{t+1}) :

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_{\mathcal{D}} \left[\text{MSE}(\hat{z}_{t+1}, z_{t+1}^{\text{target}}) + \beta_J \text{KL}(q_{\phi}(z \mid o_t, g_t) \parallel \mathcal{N}(0, I)) \right]. \quad (3)$$

The replay data $\mathcal{D}$ contains recorded transitions. The first term measures next-embedding prediction error. The second regularises the encoder’s Gaussian distribution $q_{\phi} = \mathcal{N}(\mu_{\phi}, \text{diag } \sigma_{\phi}^2)$ against a standard normal reference; β_J controls its weight. This is *not* a KL divergence against an LSTM-predicted next-state prior. Nor is it an H -step training loss: repeated rollouts are a planning operation, while this objective supervises one step at a time. Gaussian regularisation and stop-gradient define the current training choice; neither alone establishes that representations cannot collapse or that they retain all strategically relevant information.

How predictions become decisions. Planning starts with a candidate legal action and repeatedly applies a latent transition model to imagine a continuation. Predicted rewards or values score those continuations; their scores are aggregated to compare candidate first actions. A generic predicted-return estimate is

$$\hat{J}_H(a) = \frac{1}{K} \sum_{k=1}^K \sum_{\ell=1}^H \gamma^{\ell-1} \hat{r}_{t+\ell}^{(k)} \quad \text{with first action } a_t = a. \quad (4)$$

Here H is planning depth, K the number of sampled continuations, γ a discount factor and $\hat{r}$ a predicted reward. Larger values are preferred. No complete successor state is generated and the rules engine is not called inside these imagined continuations. The real first action must still come from the engine’s legal set.

JEPA prediction, reward/value learning and search are separate components. The present dream-search path uses recurrent stochastic dynamics and a predicted-reward head; training the JEPA head does not by itself replace that rollout model. A future JEPA-driven planner must qualify this connection explicitly. Multi-step error, latent collapse, reward accuracy and the legality of imagined continuations therefore remain distinct evaluation questions; a low one-step embedding loss is not evidence of accurate long-horizon play.

3.3 Active inference as path-integral control

We first distinguish *belief estimation* from *action selection*. In the notation of Section 3, s is the hidden game state and o an observation. A generative model $p_{\text{gen}}(o, s)$ describes how states produce observations; an approximate distribution $q_{\text{bel}}(s)$ represents the agent’s current belief. Friston’s variational free energy (Friston, 2010) is

$$F[q_{\text{bel}}] = \text{KL}(q_{\text{bel}}(s) \parallel p_{\text{gen}}(s \mid o)) - \log p_{\text{gen}}(o). \quad (5)$$

For a fixed observation the evidence term is constant, so minimising F makes the approximate belief closer to the model’s posterior. It need not make that posterior less uncertain. This estimation problem does not yet specify which action to take.

Goals, defaults and controlled predictions. Let η denote the game-state coordinates on which goals are evaluated, such as survival, opponent elimination and material advantage. These are properties of a possible game position, not the learned JEPA embedding. A preference distribution $p_{\text{pref}}(\eta)$ specifies which positions are desirable, with cost

$$V(\eta) := -\log p_{\text{pref}}(\eta). \quad (6)$$

Lower V means a more preferred position. One may specify costs directly and read $p_{\text{pref}} \propto e^{-V}$, up to normalisation; the costs are not calibrated win probabilities.

A policy π chooses actions from beliefs. Let τ denote a horizon-length trajectory, including states, actions and observations. $Q^\pi(\tau)$ is its predicted distribution when following π , accounting for hidden-state uncertainty and future observations. A default policy $q_0(a \mid b)$ instead induces the reference trajectory distribution $P_0(\tau)$. This *passive prior* is a baseline way of acting, not the agent’s belief about hidden cards; “passive” does not mean that no actions occur.

What a Gibbs tilt means. Following Kaufmann (2026), define a target trajectory distribution by reweighting the default trajectories:

$$P_V(\tau) = \frac{1}{Z} P_0(\tau) e^{-V(\eta_\tau)}, \quad Z = \mathbb{E}_{P_0}[e^{-V(\eta_T)}]. \quad (7)$$

This is a *Gibbs tilt*: multiply each baseline probability by an exponential preference weight, then renormalise so probabilities sum to one. With equally probable trajectories and terminal costs 0 and 1, their weights become 1 and e^{-1} , giving approximately 0.73 and 0.27. It changes the distribution we want to approach; it does not change the rules of the game.

The control objective measures how far the policy’s predicted trajectories are from that target:

$$G(\pi) = \text{KL}(Q^\pi(\tau) \parallel P_V(\tau)). \quad (8)$$

Substituting (7) gives an interpretable decomposition:

$$G(\pi) = \underbrace{\text{KL}(Q^\pi \parallel P_0)}_{\text{deviation from default}} + \underbrace{\mathbb{E}_{Q^\pi}[V(\eta_T)]}_{\text{expected terminal cost}} + \log Z. \quad (9)$$

The constant $\log Z$ can be dropped when comparing policies. The first term discourages costly deviations from a specified reference; the second favours good game outcomes. If the controlled and reference processes share transition/observation kernels, their action-conditioned trajectory KL decomposes into expected action-policy divergences, $\sum_t \mathbb{E}[\text{KL}(\pi(\cdot \mid b_t) \parallel q_0(\cdot \mid b_t))]$. A deterministic choice of action a at one decision has cost $-\log q_0(a \mid b)$. This does not make all state-only trajectory divergences equal to action divergences: actions can lead to indistinguishable states.

Kaufmann’s continuous-time result relates the corresponding path objective to stochastic model-predictive control under its diffusion assumptions. MTG actions are discrete; we use the belief-space formulation rather than claiming the Rust engine is a continuous diffusion.

Information is valuable through subsequent decisions. Information can be valuable when it lets the agent make better subsequent decisions. A planner that accounts for these contingent decisions can favour useful information-gathering actions without adding a separate information bonus. This interaction between acting and learning is called *dual control* (Fel’dbaum, 1960). Scrying and drawing cards remain worthwhile when their gameplay benefits outweigh their costs.

For example, casting a less important spell before committing a key threat both advances the position and elicits a response. If the opponent spends a counterspell, that response changes what the agent knows and how it should play next. The information matters because it enables a better follow-up, not simply because uncertainty decreases. In *closed-loop* planning, later actions depend on what is observed; a fixed, open-loop action sequence cannot express that contingency. A short or inaccurate planner can therefore miss useful information. This describes a capability to be evaluated, not a demonstrated property of our current planner.

Why not the conventional factorisations. The two EFE forms in common use are

$$G_{\text{IG}}(\pi) = -I_Q(\eta; o \mid \pi) - \mathbb{E}_Q[\log p_{\text{obs}}(o)], \quad (10)$$

$$G_{\text{AMB}}(\pi) = \text{KL}(Q(\eta \mid \pi) \parallel \tilde{p}_{\text{pref}}(\eta)) + \mathbb{E}_{Q(\eta \mid \pi)}[H[p_{\text{obs model}}(o \mid \eta)]]. \quad (11)$$

Here I_Q is expected mutual information (reduction in hidden-state uncertainty), H is Shannon entropy, p_{obs} is a preference over observations, $\tilde{p}_{\text{pref}}$ a state preference, and $p_{\text{obs model}}$ the observation likelihood. Observation preferences describe desired readings; likelihoods describe how readings arise. The factorisations have distinct assumptions and should not be treated as interchangeable objectives (Millidge et al., 2021; Kaufmann, 2026).

- **Information without instrumental value (“noisy TV”).** If uncertainty reduction earns an unconditional bonus, an agent can prefer learning facts that do not improve its later decisions. A stylised MTG test would offer repeated library inspection when the information cannot change any useful continuation, while using mana or forfeiting tempo needed to stabilise. Scry, surveil and cantrips are not inherently pathological: they can improve draws, supply cards or enable synergies. A failure is choosing them for the bonus *despite* their total effect on the game objective. Moreover, genuinely independent random noise has zero mutual information with the hidden state; a correctly calculated information-gain term need not reward it. Capture requires, for example, learnable but irrelevant variables or a misspecified novelty/uncertainty proxy. The test must distinguish these cases.
- **Avoidance of uncertain observations (“scotophobia”).** The term means an aversion to “dark”, poorly observed situations. The ambiguity term in (11) penalises entropy of observations *conditional on a state*, not simply the variance of game rewards or uncertainty about an opponent’s hand. Confusing these quantities can make informative or tactically favourable lines look costly merely because their feedback is uncertain. As a toy cost comparison, a line giving terminal costs -4 with probability 0.75 and 2 with probability 0.25 has expected cost -2.5 , better than a safe line costing -1 . With equal action costs, an additional uncertainty surcharge above 1.5 reverses that ordering. Whether such a surcharge actually arises from the chosen observation model must be tested; unknown blockers or open mana alone do not establish it. Avoiding a risky line may instead be rational under loss-sensitive preferences.

These are **testable failure hypotheses, not observed MTG results**. The legacy ablation selectors use simplified entropy/likelihood proxies, not a validated exact implementation of every observation-channel factorisation. Section 6.2 must establish both the proxy’s meaning and the behavioural effect before attributing a game outcome to either mechanism.

Soft selection rather than a hard minimum. On a finite candidate set, let $q_0(\pi)$ be reference probabilities and $J(\pi)$ the predicted task cost of each candidate, before adding the KL regularisation for this selection step. Minimising $\mathbb{E}_r[J] + \alpha^{-1} \text{KL}(r \parallel q_0)$ over selection distributions r yields

$$p_\alpha(\pi) = \frac{q_0(\pi)e^{-\alpha J(\pi)}}{\sum_j q_0(\pi_j)e^{-\alpha J(\pi_j)}}, \quad \text{softmax}_\alpha(J) = -\frac{1}{\alpha} \log \sum_i q_0(\pi_i)e^{-\alpha J(\pi_i)}. \quad (12)$$

This is again exponential reweighting. For equal priors, costs 0 and 1 and $\alpha = 2$, the lower-cost option receives probability approximately 0.88. As $\alpha \rightarrow 0$ selection approaches the prior; as $\alpha \rightarrow \infty$ it concentrates on minimum-cost candidates within the prior’s support. The scalar α is inverse temperature (selection precision), distinct from JEPA’s β_J . It is not itself a measured compute budget: rollout count, horizon and wall-clock limits control computation separately. Nor should the same deviation cost be included in J and charged a second time through the KL regulariser.

Where other signals belong. Heuristic, graph and language-model scores can shape the reference action policy q_0 : for example, a known combo completion may become a more probable default. The controller then evaluates whether predicted consequences justify following or departing from that default. Prior design still involves choices and calibration; replacing a weighted score sum with KL regularisation does not make these assumptions disappear. Below, the shorter symbol q denotes this action prior, never q_{bel} or the encoder distribution q_ϕ . Current native KL pilots use analytic state projections and forecasts, not a trained JEPA transition model, and do not qualify the complete learning architecture.

3.4 Knowledge graphs as a symbolic prior

Cards form an inductive graph $\mathcal{G} = (V, E, \tau)$ with typed edges $\tau \in \{\text{combo}, \text{synergy}, \text{counter}, \text{enables}\}$ ingested from the OWL ontology and an embedding pipeline trained with GraphSAGE (Hamilton et al., 2017). GraphSAGE is inductive: a freshly printed card v_{new} obtains an embedding $\mathbf{h}_{v_{\text{new}}} \in \mathbb{R}^{128}$ by sampling and aggregating its neighbours, without retraining. We exploit this in two ways: (i) the encoder feeds a graph latent $g_t = \text{Enc}_{KG}(\mathcal{G}, s_t)$ into the recurrent dynamics so that the world model’s prior is graph-aware, and (ii) the passive prior q of (9) is tilted by graph distance to combo completions, mirroring GraphRAG’s “relational evidence” pattern (Edge et al., 2024).

In implementation, we separate immutable source facts (card oracle data, ontology-grounded edges) from learned extension evidence. Self-play enrichment appends provenance-linked learned nodes/edges instead of mutating base facts, allowing many agents and runs to contribute to one shared strategic memory while preserving reproducibility and auditability. Section 4.2 makes this separation a property of the schema rather than a convention.

3.5 The neuro-symbolic boundary

“Neuro-symbolic” is used loosely enough that it is worth stating exactly what is symbolic here, what is learned, and what crosses between them. The system contains three symbolic artefacts and three learned ones, and the interfaces between them are narrow by construction.

Symbolic side. (i) The **phase-rs** rules engine is a verified, deterministic transition system: it decides legality, resolves the stack, applies state-based actions, and is the only component permitted to mutate game state. (ii) The OWL 2 DL ontology (Section 4.2) fixes the vocabulary in which card semantics, turn structure, roles, and strategic constructs are expressed, and is checked by a reasoner and by SHACL shapes. (iii) The knowledge graph is the ABox populated against that TBox from Scryfall and combo databases.

Learned side. (i) A GraphSAGE encoder turns the graph into dense node embeddings. (ii) The JEPA world model learns latent dynamics from self-play trajectories. (iii) Language models supply a prior over the legal-action set.

What crosses, and in which direction. The crossings are the substance of the architecture:

1. **Symbolic $\rightarrow$ neural (structure as inductive bias).** Graph structure is compiled into the encoder’s context channel g_t , and the legal-action set produced by the engine masks the controller’s output. The neural components therefore never have to learn legality or card identity; they learn only what is genuinely statistical.
2. **Symbolic $\rightarrow$ neural (structure as reference measure).** The knowledge graph and the heuristic shape the passive prior q of (9). Symbolic knowledge thus enters the control objective as a *cost of deviation*, not as an additive score.
3. **Neural $\rightarrow$ symbolic (induction of new relations).** Self-play traces are mined into candidate relations — card co-occurrence in winning positions, answer/threat pairs, archetype signatures — and written back as *induced* assertions. This is the direction most neuro-symbolic systems leave empty, and it is where the interesting failure modes live: a statistical regularity is not an ontological commitment.
4. **Validation as the gate between them.** Nothing induced is promoted to curated status automatically. Candidate relations are reified, weighted, attributed to the run that produced them, and validated by the shapes and reasoner before a human may accept them. This mirrors the finding of [Tiwari et al. \(2026\)](#) that the validator, not the generator, is what turns diverse neural proposals into a high-precision graph.

Curated versus induced semantics. The design commitment underlying all four crossings is that deterministic facts and learned regularities must remain distinguishable forever. A card’s oracle text is curated: it is definitional, immutable, and wrong is wrong. That two cards co-occur in 83 % of the games one of them wins is induced: it is defeasible, provenance-bearing, and may be an artefact of the deck pool sampled. Collapsing the two — writing a learned edge into the same relation as a curated one — destroys the ability to recover ground truth and makes the graph unauditable. Section 4.2 enforces the distinction with an `mtg:epistemicStatus` annotation on every axiom and a schema-level requirement that no induced assertion may exist without a `prov:Activity` that generated it.

3.6 General LLMs as indexed-action selectors

The action interface supplies a filtered observation o_t and an indexed engine-authored action list $\mathcal{A}_t$. A general chat LLM is asked to select an index:

$$\hat{i} = \text{LM}(\rho(o_t, \mathcal{A}_t)), \quad a_t = \mathcal{A}_t[\hat{i}]. \quad (13)$$

Only an index resolved to an original legal action can reach the engine. This restricts the action interface, not necessarily the model’s token generation: the current chat path parses a text reply rather than exposing a token-masked, calibrated probability distribution. Malformed replies and service failures need explicit handling and accounting. A low sampling temperature reduces output variability; it neither proves calibration nor represents a uniform-fallback probability.

3.7 Decision-tuned models as option selectors

A *decision model* here is a pretrained language model adapted to choose among supplied alternatives. It is not a world model: it returns a choice, not a predicted successor state or a simulated trajectory. Unlike a general chat selector prompted to emit an integer, its training and API are specialised for categorical decisions. Tev1 is an experimental Qwen3.5-derived family with supervised decision adaptation and a retained language-model output head ([Together AI, 2026](#); [Ollama, 2026](#)); decision tuning does not by itself establish MTG expertise.

Given a filtered observation o_t , a context projection $R(o_t)$ and a subset $\mathcal{S}_t \subseteq \mathcal{A}_t$ of engine-authored actions, it supplies

$$d_\psi(a \mid R(o_t), \mathcal{S}_t), \quad a \in \mathcal{S}_t, \quad (14)$$

where ψ denotes the selector parameters. The selected option maps back to the original action, preserving targets and object identity. Returned option probabilities indicate the model’s preference among the presented alternatives, not a probability of winning or a calibrated probability that a choice is correct.

The candidate generator is part of the policy: if a winning line is absent from $\mathcal{S}_t$, no reranker can recover it. Likewise, projection R can omit strategically important context. For example, selecting removal from a shortlist can be sensible only if the supplied state distinguishes the relevant threat, available mana and target. Measuring legal output is therefore insufficient; matched-context/candidate comparisons, shortlist coverage, tactical quality and inference cost are separate measurements (Section 6.1).

There are three possible uses, which should not be conflated: direct action selection, reranking candidates proposed by a planner, and supplying probabilities to the planner’s passive prior q_0 . Only the first is currently integrated for Tev1. None replaces the rules engine or provides the latent transition model needed for JEPA rollouts. The existing local API smoke establishes interface compatibility; full-game inference still encounters context-budget limits, and no Tev1 playing-strength result is claimed.

3.8 Self-play promotion as fixed-point iteration

Training combines all three signals. The replay buffer $\mathcal{D}$ is collected by self-play between agent variants; the world model is fit on $\mathcal{D}$; and the controller is improved iteratively by the champion-vs-challenger schema implemented in `scripts/train_pipeline.py`. At each iteration i , a challenger π_{i+1} is trained against π_i on the latest replay window and faces the champion in a best-of- N match; promotion happens iff the empirical win rate exceeds 0.55. This is a stochastic-approximation fixed-point operator on the policy improvement step (Tesauro, 1995; Silver et al., 2017, 2018; Schrittwieser et al., 2020): under standard Robbins–Monro conditions it converges, in expectation, to a fixed point of the Bellman optimality operator restricted to the dynamics class $\hat{T}$ that the world model has learned. Convergence in practice depends on $\hat{T}$ being locally accurate, which we monitor by tracking JEPA validation loss across iterations.

4 System Architecture

4.1 Game engine

The runtime engine is `phase-rs` (Rust) (phase-rs contributors, 2026), published at <https://github.com/phase-rs/phase> and accessed over its WebSocket interface via `src/integrations/phase_rs/`. Our Python side is intentionally adapter-only at execution time: `client.py` handles protocol envelopes, `runner.py` drives game loops, `adapter.py` maps legal actions/snapshots into our policy API, and `server_process.py` manages local server lifecycle for training scripts.

Tracking a living upstream. `phase-rs` is under active development and its wire protocol is versioned and enforced at handshake. Over the period covered by this report the upstream protocol advanced from v6 through v75 to the current v106 pin, which is a useful reminder that an external rules engine is a dependency with its own release cadence, not a fixed substrate. Two of those changes were structural rather than additive and required adapter work: rejections became a typed `ActionRejection` carrying a stable machine-readable code and a recovery disposition rather than a prose string, and the engine now publishes *viewer interactions* — an

Comprehensive Rules concept	CR section	Implementing module
Turn structure / phases / steps	500–514	<code>src/engine/phases.py</code>
APNAP priority & priority loop	116–117	<code>src/engine/priority_loop.py</code>
The stack & spell resolution	405–608	<code>src/engine/stack.py</code>
Layer system (continuous effects)	613	<code>src/engine/continuous_effects.py</code>
Replacement effects	614–616	<code>src/engine/replacement_effects.py</code>
Triggered abilities	603	<code>src/engine/triggered_abilities.py</code>
State-based actions	704	<code>src/engine/state_based_actions.py</code>
Combat (declare, block, damage steps)	506–510	<code>src/engine/combat.py</code>
Mana & cost payment	106, 601–602	<code>src/engine/mana.py</code>
London mulligan	103.5	<code>src/engine/game_setup.py</code>
Cleanup discard to hand size	514	<code>src/engine/phases.py::cleanup_step</code>
Loss from empty library on draw	104.3c, 704.5b	<code>src/engine/state_based_actions.py</code>
Legal-action enumeration	— (engine-internal)	<code>src/engine/rules_engine.py</code>

Table 2: Mapping from Comprehensive-Rules concepts to the implementing engine module. Every cell is a real path; auditing a rule reduces to opening that file. The legal-action enumerator (`rules_engine.py`) is the only point of contact between the engine and any agent.

explicit, engine-authored protocol for the choices a player must make during resolution (targets, modes, damage assignment, ordering). The second is the more significant for us: it moves a large class of decisions out of the flat `legal_actions` list into a structured opportunity schema, which is strictly better for an agent that wants to reason about *what kind* of choice it is making. We pin the protocol constant in `client.py` and fail closed on mismatch rather than attempting to negotiate, on the principle that a silently misinterpreted game state is worse than a refused connection.

Core integration modules:

- `client.py`: protocol handshake, message parsing, and typed message dataclasses for `GameStarted`, `StateUpdate`, and `GameOver`;
- `runner.py`: end-to-end game loop for a Python-controlled seat, including structured per-decision JSONL traces;
- `adapter.py`: practical `GameAction` ↔ `Action` bridge and snapshot projection for native Python policies;
- `agent_bridge.py`: random/heuristic/LLM pickers plus adapter-backed native-agent picker support;
- `server_process.py`: managed local `phase-server` process startup/shutdown for reproducible training/evaluation runs.

The old Python engine remains in-repository as legacy compatibility code, but new experiments and reported benchmarks use `phase-rs` as the authoritative runtime transition system.

Table 2 maps each Comprehensive-Rules concept to the exact module that implements it; readers reproducing the system can use it as a reading order through the codebase.

4.2 Symbolic substrate: the ontology

The symbolic layer is an OWL ontology of MTG derived from the Comprehensive Rules and aligned to the DOLCE-UltraLite foundational ontology, with OWL 2 DL as its modelling target. It is a contribution in its own right and is described in a companion paper (Neubürger and

Neubürger, 2026); we summarise here the intended symbolic contract and its current qualification boundaries.

Three properties matter for the argument of this paper. First, the ontology is *expressive enough to be wrong*: logical analysis exposed incompatible property characteristics that were repaired, and Hermit finds the import-resolved schema consistent with no unsatisfiable named classes. This is not complete OWL 2 DL profile conformance of the raw external documents, nor certification of a populated card graph. The CR 205.3 card-type/subtype correlation is also checked procedurally. Second, its factual vocabulary is *derived, not curated by hand*: 3,165 source rules are parsed and 830 vocabulary subjects carry direct document provenance; rule text remains local rather than being redistributed. Third, and most importantly here, it distinguishes *curated* from *induced* assertions at the schema level. That distinction supplies a necessary audit boundary for the learning loop of Section 4.3.

For orientation: cards are imported from Scryfall as `CardDesign` nodes with oracle text and an SBERT embedding; mechanics, abilities, archetypes and combos are first-class nodes; APOC supplies PageRank, community detection and shortest-path queries; and multi-hop retrieval over this structure is the “relational evidence” pattern of GraphRAG (Edge et al., 2024), which suits questions — *which two cards in my library combo, and does either die to a counterspell?* — that similarity search answers badly.

4.3 Playing to build the graph

The knowledge graph is usually treated as a fixed input: someone curates it, the agent reads it. We are interested in the other direction as well, because an agent that plays thousands of games observes something a curator cannot — which cards actually co-occur in positions that get won, against which opponents, at which points on the curve.

The loop has four steps. Agents play on the rules engine and emit structured decision traces. Traces are post-processed into candidate relations: card pairs recurring in winning positions, answer/threat pairs, archetype signatures. Candidates are written into the graph as *induced* assertions — reified, weighted by support and refutation counts, and attributed under PROV-O to the run that produced them. Later agents query those assertions as an additional prior, and their play produces the next round of evidence.

Two design commitments keep this from degenerating.

The extension layer is append-only and typed apart. An induced synergy is not an edge in the curated `synergisesWith` relation; it is a reified evidence node that the schema requires to be attributable to an activity. A consumer needing deterministic ground truth recovers it by filtering on epistemic status. This matters because the two kinds of claim have different logic: a card’s oracle text is definitional and wrong is wrong, whereas a co-occurrence statistic is defeasible and may be an artefact of the deck pool sampled. Systems that write learned edges into curated relations lose the ability to tell the difference, and with it the ability to audit.

Promotion is not automatic. Nothing is reclassified from induced to curated without a human. We regard the promotion gate — what support threshold, what handling of disagreement between runs (Ehrenmüller et al., 2026) — as an open problem rather than a solved one, and say so in Section 7.

Agent reasoning is itself in the graph. A decision is recorded as a `prov:Activity` carrying the objective decomposition it was selected by, the belief it was taken under, and the policy precision in force. “Why did seat 0 attack on turn 6?” is therefore a graph query rather than a log grep. This is a deliberate response to the observation (Sabou, 2026) that trustworthy neuro-symbolic systems need explanations grounded in domain knowledge with provenance

attached, rather than post-hoc attribution scores; it also means the explanation and the game state it refers to are expressed in one vocabulary.

Collective intelligence, stated carefully. The attractive reading is that heterogeneous agents accumulate shared strategic memory across runs, so that later agents inherit what earlier ones discovered without retraining. The framework supports that reading structurally. We have not yet demonstrated it: showing that graph enrichment from run n measurably improves agents in run $n+1$, and is not merely overfitting to a fixed deck pool, is the experiment that would substantiate the claim, and it is not among the results reported here.

4.4 World model

Let o_t be the perspective-filtered observation at decision step t and g the player’s KG context (deck, archetype tag, known opponent information). The world model decomposes as:

$$z_t = E_\phi(o_t, g_t) \quad (\text{observation encoder}) \quad (15)$$

$$g_t = \text{Enc}_{\text{KG}}(g, o_t) \quad (\text{symbolic context encoder}) \quad (16)$$

$$z_{t+1}^M, h_{t+1} \sim M(z_t, a_t, h_t) \quad (\text{recurrent stochastic dynamics}) \quad (17)$$

$$\hat{z}_{t+1} = f_\theta(z_t, a_t) \quad (\text{JEPA predictor head}) \quad (18)$$

$$a_t = C(z_t, g_t, h_t) \quad (\text{controller}). \quad (19)$$

E_ϕ produces a 256-dimensional representation of the visible position and optional symbolic context. The JEPA objective is (3): next-embedding prediction plus encoder regularisation against a standard normal distribution. The recurrent mixture-density dynamics, reward head and controller are separate from that predictor; their use in dream search does not imply an already-qualified JEPA-driven planner (Section 3.2).

The prediction residual $\|\hat{z}_{t+1} - z_{t+1}\|_2$ measures disagreement in the chosen representation. It may flag unfamiliar transitions for analysis, but does not distinguish model error, hidden-state uncertainty and useful tactical novelty by itself. In particular, it is not an intrinsic reward in the belief-space KL objective. The encoder also supports a separate feature-reconstruction training path; this does not make reconstruction of a complete successor state part of the JEPA loss.

4.5 Combining selectors with a planner

The intended integration separates *what is a plausible default action* from *what its predicted consequences are*. Heuristic, graph or model preferences can supply $q_0(a \mid b_t)$; a transition model and an outcome cost supply the planning component of (9). The earlier weighted-fusion policy is retained as a separate baseline, not identified with the KL controller. Language-model scores must be calibrated and traced if used as probabilities, and a deck’s preferred curve is a strategic heuristic, not a hard legality constraint.

Illustrative trade-off. Suppose a prior gives probabilities 0.6 to countering a spell and 0.4 to passing. Their deterministic one-step deviation costs are $-\log 0.6 \approx 0.51$ and $-\log 0.4 \approx 0.92$. If predicted outcome costs are 0.8 and 0, the totals are approximately 1.31 and 0.92: the predicted advantage of passing overrides the prior’s preference for countering. These invented numbers illustrate the decomposition, not model outputs from a recorded game. Obtaining reliable outcome forecasts and context-conditioned priors is the experiment, rather than a consequence of writing down the objective.

4.6 Self-play training loop

We use a champion-vs-challenger promotion scheme:

Algorithm 1 Champion-vs-Challenger Self-Play (Phase A.5).

```

1: Initialise pool  $\mathcal{P}$  with  $K$  randomly-seeded agents.
2: Pick champion  $c^* \leftarrow \arg \max_{c \in \mathcal{P}} \text{ELO}(c)$ .
3: for round = 1, ...,  $R$  do
4:   Sample challenger  $c \sim \mathcal{P} \setminus \{c^*\}$ .
5:   Play  $N$  games  $c$  vs  $c^*$ , collect trajectories  $\tau$ .
6:   Update ELO for both; train all pool members on  $\tau$ .
7:    $w_r \leftarrow$  challenger win rate over the  $N$  games.
8:   if  $w_r \geq \theta$  then
9:     Promote:  $c^* \leftarrow c$ ; bump ELO by +50, demote former by -20.
10:  end if
11: end for

```

θ is the promotion threshold (default 0.55). The loop is implemented in `src/training/r1_trainer.py`; `scripts/train_pipeline.py` is the execution entry point, not evidence of a qualified end-to-end native promotion experiment.

4.7 Agent zoo: algorithmic specification

The framework combines Python agents satisfying `decide_action(state, legal_actions) → action` with native `phase-rs` pickers selecting an index into the engine’s legal-action list. They span the spectrum from combinatorial baselines to the V+M+C+JEPA+KG+LLM fusion of Section 4.4; the native pickers additionally expose KL control and decision-tuned language models. We summarise their algorithmic roles below; implementation detail belongs in the companion technical report.

Random (π^{rand}). A weighted sampler over legal actions with fixed type-class weights $w_{\text{land}} = 25$, $w_{\text{cast}} = 10$ (boosted to 30 for casts from the command zone), $w_{\text{atk}} = 8$, $w_{\text{pass}} = 1$. Used as a baseline floor; not deterministic per-instance because it shares the process-global RNG.

Heuristic (π^{heur}). Deterministic priority cascade

$$a^* = \arg \max_{a \in \mathcal{A}_{\text{legal}}} \text{prio}(a), \quad \text{prio} : \begin{cases} \text{PLAY_LAND} \mapsto 100 \\ \text{CAST_SPELL} \mapsto 90 \\ \text{DECLARE_ATTACKERS} \mapsto 80 \\ \text{DECLARE_BLOCKERS} \mapsto 70 \\ \text{ACTIVATE_ABILITY} \mapsto 60 \\ \text{PASS} \mapsto 1 \\ \text{CONCEDE} \mapsto 0, \end{cases}$$

with three tactical refinements: (i) counter-spell discipline filters `counter target` casts unless an opponent spell tops the stack, (ii) when `prefer_aggressive` is set the agent always picks an attack at the top tier, (iii) on the blocking step it chump-blocks the largest unblocked attacker with the smallest blocker. Tie-breaking uses a seeded `random.Random`, making it the strongest fully-reproducible baseline.

KG-Heuristic ($\pi^{\text{KG+heur}}$). Inherits π^{heur} but, when more than one `CAST_SPELL` is legal, re-ranks them through the knowledge graph. Let H be the agent’s hand and B its battlefield. For each candidate cast c ,

$$\text{score}_{\text{KG}}(c) = 10 \cdot \mathbf{1}[c \in \text{NearComboMissingPieces}(H \cup B)] + \sum_{p \in B} \text{strength}(\text{KG-edge}(c, p)),$$

where the indicator is computed by a single Cypher round-trip (`kg.detect_near_combos`) and synergy strength comes from the `HAS_SYNERGY` edges of the knowledge graph. The best-scoring cast replaces the cast subset in the legal-action list and the parent priority logic resumes, so `PLAY_LAND` can still beat `CAST_SPELL`. On any KG exception the agent latches `kg_disabled = True` and silently degrades to π^{heur} .

Ollama LLM (π^{LLM}). The LLM is consulted as a discrete posterior over the legal-action set, never as a free-form generator. Concretely, for each decision we serialise the board state s_t (life totals, hand sizes, lands, creatures) and an enumerated list $\{(i, a_i)\}_{i=0}^{|\mathcal{A}_{\text{legal}}|-1}$, prompt the model to “respond with ONLY the option number”, parse an integer $\hat{i}$ from the reply, and return $a_{\hat{i}}$. Formally,

$$\hat{i} = \pi_{\text{LLM}}(\text{prompt}_t), \quad a^* = a_{\text{clip}(\hat{i}, 0, |\mathcal{A}_{\text{legal}}|-1)}.$$

Illegal actions are unrepresentable by construction. We use Gemma 4-edge (`gemma4:e2b`) at temperature 0.2 via Ollama for a $\sim 1\text{--}4$ s budget per decision. On any HTTP or parse failure the agent falls back to π^{rand} and records the event in a per-game stats dictionary.

Decision-model deployment (π^{dec}). The conceptual distinction from chat selection and latent dynamics is given in Section 3.7. The current deployment uses Tev1 0.8 B/4 B through Ollama’s structured option-selection interface.

For a filtered observation x_t , let $\tilde{x}_t = R(x_t)$ be its decision context projection, $\mathcal{A}_t$ the engine-authored legal-action set and $\mathcal{S}_t = C_k(x_t, \mathcal{A}_t)$ a candidate set, $|\mathcal{S}_t| \leq k \leq 24$. The selector supplies a categorical preference

$$p_{\theta}(a \mid \tilde{x}_t, \mathcal{S}_t), \quad a \in \mathcal{S}_t,$$

and its selected key is mapped back to the *original* engine action, including targets and object identities. The first implementation uses heuristic action tiers with seeded tie-breaking, excludes concessions when alternatives exist, and retains a pass option if present. A forced choice bypasses inference. The 24-option cap follows the model’s reported training range, rather than implying that MTG’s action space has that size. We log the candidate indices, model probabilities and $|\mathcal{S}_t|/|\mathcal{A}_t^{\text{eligible}}|$. This fraction measures *action coverage*, not recall of the best move. Inference failures are explicit incomplete runs; no random-policy fallback is credited to the decision model.

Input budgeting is also an interface constraint. Local Ollama 0.35.1 rejects request bodies above 64 KiB and decision prompts above 2050 tokens. Raw engine snapshots exceed the body limit. Our versioned card-aware projection retains visible identities and instance IDs, public characteristics, life/mana and decision state, grouping equivalent card rows without merging instance identities. It omits hidden placeholders, library order, internal journals and compiled rule syntax trees; it is not a lossless rules-state encoding. Projection version and request bytes are recorded, and oversize inputs fail explicitly rather than being silently cropped. Compact full-game pilots still hit the token ceiling; a budgeted context policy must be evaluated before strength measurements.

This component can replace a chat selector at the action boundary without replacing the rules engine, the world model or KL planning. Using it as a critic over world-model candidates, or as a learned passive prior, is a separate proposed integration. Our hypothesis is improved selection reliability and a favourable strength–latency trade-off, not that decision tuning already makes Tev1 a stronger MTG player.

Active Inference / KL control (π^{KL}). Maintains a categorical belief b_t over hidden-information hypotheses about the opponent’s hand (`holds_removal`, `holds_counterspell`, `holds_threat`, `holds_nothing`). Control costs are charged against a passive prior, while analytic forecasts supply outcome costs. Two planners are available: a belief-tree reference planner and a sampled MPPI-style approximation. Current sampled continuations use averaged stage potentials and soft aggregation, not an exact calculation of the full trajectory objective. Defaults are $H = 3$, $K = 48$ and selection precision $\alpha = 4$ (called `beta` in configuration). Belief entropy is reported for diagnostics and is not an additive reward; this excludes that particular bonus mechanism, not every possible model-induced failure.

Preference potential. Goals use $V(\eta) = -\log p_{\text{pref}}(\eta)$ from (6), with costs for life/material differences, turn pressure and terminal outcomes. These coordinates make the goal specification inspectable, but its hand-set weights and incomplete state projection may still misvalue a position. Anchoring goals to game consequences avoids explicitly rewarding sensor readings; it is not a universal correctness guarantee.

World-Model agent (π^{WM}). Plays through the V+M+C+JEPA stack of Section 4.4. Every turn,

$$z_t = E_\phi(o_t, g_t), \quad g_t = \text{Enc}_{\text{KG}}(g, o_t), \quad h_t \text{ retains recurrent history.}$$

Two action-selection modes are available: a fast direct policy over the supplied legal actions, and dream search that compares predicted cumulative rewards from recurrent latent continuations, as in (4). The JEPA head is trained separately; selecting a JEPA-trained checkpoint does not establish that this search unrolls the JEPA predictor. Imagined continuations are approximate and do not receive fresh engine legality checks.

LLM-Fusion agent (π^{fuse}). A weighted-fusion baseline combining four scalar signals per legal action,

$$\text{score}(a) = \alpha_h s_{\text{heur}}(a) + \alpha_g s_{\text{KG}}(a) + \alpha_w s_{\text{WM}}(a) + \alpha_\ell \log p_{\text{LLM}}(a), \quad a^* = \arg \max_a \text{score}(a),$$

with default weights $(\alpha_h, \alpha_g, \alpha_w, \alpha_\ell) = (0.15, 0.15, 0.40, 0.30)$. The LLM signal is skipped when one action dominates the heuristic-plus-WM signal by more than `obvious_threshold` = 0.8, which saves roughly 80% of LLM calls in practice. Setting any single weight to zero recovers the corresponding ablation; the canonical four-arm study (LLM-only, WM-only, KG-only, all four) is wired up by `scripts/run_matchups.py`.

Algorithm 2 LLM-Fusion decision (one priority window).

Require: state s_t , legal actions $\mathcal{A}$, world model (Enc_V, M, C) , KG $\mathcal{G}$, opponent model b_t , weights $(\alpha_h, \alpha_g, \alpha_w, \alpha_\ell)$

- 1: $\mathbf{s}_h \leftarrow \text{HeuristicScores}(s_t, \mathcal{A})$ $\triangleright \sim 50 \mu\text{s}$
- 2: $\mathbf{s}_g \leftarrow \text{KGScores}(s_t, \mathcal{A})$ $\triangleright$ combo + synergy bonuses
- 3: $z_t \leftarrow \text{Enc}_V(s_t)$; $\mathbf{s}_w \leftarrow \text{DreamSearch}(z_t, h_t, \mathcal{A}; K, D)$
- 4: **if** $\text{spread}(\mathbf{s}_h + \mathbf{s}_w) > \tau_{\text{obvious}}$ **then**
- 5: $\mathbf{s}_\ell \leftarrow \mathbf{0}$ $\triangleright$ skip the LLM call
- 6: **else**
- 7: $\mathbf{s}_\ell \leftarrow \log p_{\text{LLM}}(\cdot \mid \text{prompt}(s_t, \mathcal{A}))$
- 8: **end if**
- 9: **return** $a^* \leftarrow \mathcal{A}[\arg \max_i (\alpha_h \mathbf{s}_{h,i} + \alpha_g \mathbf{s}_{g,i} + \alpha_w \mathbf{s}_{w,i} + \alpha_\ell \mathbf{s}_{\ell,i})]$

Reasoning traces. Every agent writes a `ReasoningTrace` dict to `self.last_reasoning` before returning. The orchestrator flushes these to `runs/<out>/logs/game_NNNN.jsonl`, one line per decision, schema {turn, phase, player_id, agent_kind, rationale, legal_action_count, chosen_index, top_candidates, scores, beliefs}. This is the canonical telemetry format consumed by the post-hoc analysis notebooks and by the Streamlit visualisation panel discussed in Section 8.

5 Implementation

We keep implementation detail intentionally brief here because the scientific object of interest is the *interaction* between symbolic structure and learned planning, not the package layout of the repository. The released artifact instantiates four coupled layers: a rules-correct simulator, an ontology-backed card graph, a latent world model trained from self-play trajectories, and an agent layer that can combine heuristic, graph, world-model, and LLM signals. The code is modular enough that each layer can be ablated independently, but the paper’s main claim is about the resulting research design: MTG becomes a controlled setting for studying how symbolic priors, learned dynamics, and uncertainty-aware action selection cooperate.

Reproducibility. The current `phase-rs` benchmark records the repository and engine checkout revisions, the actual server identity from the handshake, protocol version, deck fingerprint, full configuration and per-game picker seed. Candidate selection uses a seeded Python RNG. Engine shuffles, opponent-AI randomness and model-service inference are not all controlled by that seed; matched picker seeds are therefore not deterministic paired-game replays. Decklists are committed under `data/decks/`. Engine RNG control and seat rotation remain required for a fully paired comparison.

6 Preliminary Evaluation

We report preliminary results; a full benchmark suite (Section 8) is in development. The numerical results below are archived experiments from the earlier research cycle, not outcomes of the new protocol-v106 or Tev1 benchmark. The local Tev1 API smoke verifies interface compatibility only; no MTG playing-strength result for Tev1 is reported here.

6.1 Decision-model benchmark protocol

What is being tested. The unit of deployment is one controlled seat against a fixed `phase-ai` difficulty, deck and rules-engine revision. We compare heuristic selection, general chat-model selection, Tev1 0.8B/4B and belief-space KL control. The question is whether decision adaptation improves legal-action selection reliability and playing strength per unit of inference cost. Legal output alone is not evidence of tactical quality. The available episode harness is a built-in-AI opponent benchmark, not yet an agent-vs-agent round robin or a four-player Commander evaluation.

Measurements and failure accounting. The implemented harness reports completed-game win rate with Wilson 95% intervals, completion rate and terminal-reason counts. Genuine engine-terminal draws are distinguished from timeouts, action caps, rejections, AI-driver faults and inference errors. A run with no completed games has an undefined win rate, not a zero score. Reporting the completion denominator alongside win rate is necessary because dropping difficult failed games can bias apparent strength. Decision traces record latency; aggregate p50/p95 includes attempted decisions in the retained game trace, including explicit inference failures. Candidate coverage and the number of decisions requiring a shortlist are also reported. Forced choices and warm/cold model starts must be separated in the eventual performance study.

Comparison	Scientific question	Required controls
Deployment stacks	Which complete policy offers the best strength–cost trade-off?	Fixed engine, decks, difficulty, hardware; disclose candidate and context differences.
Matched rerankers	Does decision tuning improve choice quality?	Same saved observations, candidate identities, descriptions and option order; size-matched Qwen3.5 chat/base comparator.
Shortlist ablation	How much does the candidate generator determine the result?	Vary k and candidate source; report coverage, action-type omissions and order sensitivity.
Planner integration	Does the selector help world-model candidates or KL priors?	Hold rollout model, horizon and compute budget fixed; change only the selector/prior.

Table 3: Planned decision-model comparisons. Only the deployment adapter, tracing and episode harness are implemented; the controlled reranker and planner-integration experiments are follow-up work.

What remains controlled experimentally. The pilot uses a fixed explicit deck and interleaves objective arms over the same picker-seed schedule. Replication across held-out deck archetypes, engine/AI seeds and seat assignments is planned after interface validation; we do not tune and report on the same holdout. The present chat bridge uses a compact context and permits random fallback, whereas Tev1 receives a card-aware filtered projection and surfaces inference errors. This is a *deployment-stack* comparison until context, shortlist and fallback accounting are matched. Shortlists of 8, 16 and 24 options, controlled option permutations and a size-matched general-model comparator are proposed ablations, not completed measurements. Benchmark outputs record model identifiers and configuration; runtime versions, quantisation, hardware and warm-up state must accompany published timing results.

Data freeze and Commander qualification. We maintain separate, versioned inputs: Scryfall oracle records and rulings for immutable card facts and graph construction, and the pinned engine’s compiled definitions for executable mechanics. Updating one does not update the other or retrain a policy. Evaluation must record source dates, checksums, deck payloads, engine/policy revisions and the support status of all study cards; graph population is validated against that same frozen snapshot.

The native pod wrapper can configure four Commander decks with one Python-controlled host and three native AI opponents. This is not a four-policy zoo tournament. Terminal pod qualification, elimination handling, transport limits and seat/deck rotation are separate gates; a turn-capped development smoke is never counted as a completed pod or draw. Multiplayer beliefs and diplomacy are not established by running a single-opponent latent planner in a four-player room.

Publication sequence and statistical scope. The executable first stage is a bounded feasibility pilot of runtime-qualified random, heuristic and objective stacks. It checks completion and audits action semantics before adding more arms. Decision models, matched chat rerankers, KG induction and learned JEPA/dreaming enter strength evaluation only after their interface, provenance and model-calibration gates pass. The standalone ontology resource can be evaluated independently; an early agents paper must restrict its empirical claims rather than imply that the whole proposed architecture has been measured.

Current feasibility evidence and interface repairs. An October 5 Modern burn-mirror pilot attempted ten games across five policy stacks: eight reached terminal losses, while both KL games hit the 1,500-action cap. The traces exposed equal forecasts for cancellation and casting subdecisions, rather than a protocol rejection. The analytic model now distinguishes cancellation from a one-time pending-cast completion forecast, and resolves simple fixed-damage player/object targets from the native ability representation without changing the submitted legal action. A fresh two-game KL check reaches terminal losses in seven and eleven turns (203 decisions combined; no cancellations). These checks repair a concrete failure, but do not qualify card-/mode-exact dynamics, causal objective comparisons, or playing strength.

Full-session reconnect also now submits the exact server-issued generation and authenticated player credential and checks the restored seat and revision. A forced socket-close test restored the same native session; long Commander pods remain unqualified. The oracle refresh and ontology audits are offline file operations, not proof of a successful Neo4j import. No Fuseki service is configured, and current native pilots do not demonstrate live ontology-grounded decision making. The standalone resource paper evaluates the ontology separately.

After pilot diagnostics, we predeclare comparisons and freeze a fresh holdout, including a fixed failure policy and sample size. Games are the analysis unit; the four correlated seats of a pod and its many actions are not independent replicates. We report stratum-specific effects and intervals, apply multiplicity control to confirmatory comparison families, and retain incomplete and unstarted cells separately. A 100-game binomial cell has approximately ten-percentage-point uncertainty near 50%; it cannot resolve a five-point advantage. An independent two-arm comparison near 50% needs roughly 1570 games per arm for 80% power at two-sided 5% significance to detect that effect, before additional dependence or multiplicity. Pilot durations therefore inform both a power analysis and an explicit compute budget, not an automatic multi-day run.

6.2 Archived preliminary experiments

(i) Engine correctness. The repository contains ~ 50 pytest modules (513 passing tests, 27 skipped, 0 failures) covering combat, the stack, triggered/static/replacement/continuous abilities, watchers, face-down permanents, delayed triggered abilities, and end-to-end games. The whole-suite runtime is under 20 seconds on a single CPU.

(ii) Knowledge-graph coverage. The OWL ontology v1.1 declares 17 top-level classes and 24 object/data properties. After Scryfall ingestion of a representative Standard environment ($\sim 5,000$ cards) the graph contains the cards, their oracle-text SBERT embeddings, and combo edges imported from Commander Spellbook.

(iii) Self-play promotion loop. The promotion loop of Algorithm 1 is exercised by two unit tests (`tests/test_rl_trainer.py`); both pass. End-to-end promotion behaviour — whether champions stabilise after $\mathcal{O}(10)$ rounds of $N = 20$ games — is the next planned experiment and is not reported here.

Checkpoint-benchmark status. We ran a clean checkpoint evaluation using `scripts/benchmark_trained_agents.py` across four archived JEPA checkpoints (`jepa_epoch_10.pt`, `epoch_30`, `epoch_50`, `jepa_final.pt`) against `random` and `heuristic` baselines (8 games per baseline per checkpoint, symmetric Mono-Red mirror, seed 77, `-max-turns 30`). Results are in Table 4. The final checkpoint achieves 25% win rate against both baselines, below the heuristic-vs-random parity level of $\sim 33\%$, indicating that the world model still needs more training data or epochs to match the hand-tuned heuristic. The active-inference-inspired checkpoint selector identifies `jepa_final.pt` as best by expected free energy (EFE 0.081; lower is better).

Checkpoint	vs. random WR	vs. heuristic WR	EFE
jepa_epoch_10.pt	12.5%	0.0%	0.455
jepa_epoch_30.pt	12.5%	12.5%	0.401
jepa_epoch_50.pt	25.0%	25.0%	0.105
jepa_final.pt	25.0%	25.0%	0.081

Table 4: JEPA checkpoint evaluation on symmetric Mono-Red mirror match (8 games/baseline/checkpoint, seed 77, max-turns 30). Win rate against **random** and **heuristic** baselines. EFE is the active-inference-inspired selection score (lower is better).

(iv) **Pairwise ablation against the heuristic baseline.** We use the in-tree `scripts/run_matchups.py` harness to play each candidate agent versus the strategy-aware **HeuristicAgent** on a fixed Modern matchup (Mono-Red Burn vs. Azorius Control), 16 games per cell with seat order swapped to remove going-first bias, `-max-turns 30`, seed 7. Results are summarised in Table 5. The KG-grounded heuristic and the LLM agent both clear 50% against the baseline; the world-model agent ties; the LLM-WM-KG *fusion* agent underperforms its own components, which we attribute to (a) its much larger per-decision latency biasing toward conservative passes when the cost-aware short-circuit fires, and (b) uncalibrated mixture weights ($\alpha_h, \alpha_g, \alpha_w, \alpha_\ell$) in the additive scoring of Section 4.7. We record this result because it is the direct motivation for the reformulation in Section 3.3: a weighted sum of four heterogeneous signals has four free parameters, no unit in which the terms are commensurable, and no principle for setting them. Belief-space KL control removes the choice — the knowledge-graph, heuristic and language-model signals shape the passive prior q , and the price of deviating from it is fixed at $-\log q(a)$ rather than tuned.

Agent vs. heuristic	Wins	Win-rate	Avg. turns	Avg. s/game
kg_heuristic	9 / 16	56.3%	6.4	0.32
llm (Ollama)	9 / 16	56.3%	5.9	1.02
world_model	8 / 16	50.0%	7.1	2.22
llm_fusion	5 / 16	31.3%	7.8	114.46

Table 5: Pairwise win-rate of each candidate agent versus the strategy-aware **HeuristicAgent** on a single Modern matchup (Mono-Red Burn vs. Azorius Control), 16 games per agent, seat-swapped, seed 7. Average wall time is per game on a single CPU; the LLM agent uses an Ollama-hosted Llama-3 model with `keep_alive` caching.

(v) **Multi-agent round-robin (Standard, Swiss).** We additionally ran a 7-agent Swiss tournament on the same Modern environment, 6 rounds, with Buchholz tiebreak (Table 6). The hand-tuned **HeuristicAgent** is currently undefeated; **ActiveInferenceAgent** and **LLMFusionAgent** share second place. The lower-bound **RandomAgent** sits at the expected level. The world-model and bare-LLM standings reflect the harness pre-fix — the snapshot of the corrected harness (1v1, Table 5) is the more reliable measurement, and a re-run of the multi-agent round-robin under the patched runner is queued in Section 8.

(vi) **Commander pod.** A 4-seat 8-round EDH pod (Krenko / Atraxa / Urza / Meren cores, seat rotations to remove turn-order bias) gave the standings in Table 7: the KG-grounded heuristic was the clear winner, with **LLMFusionAgent** third despite its high per-turn cost. With only 4 distinct decks the pod is too small to draw strong conclusions, but it demonstrates that the system runs Commander games to completion at the expected ~ 8 –12 turn length without crashing.

Rank	Agent	W-D-L	Pts	Buchholz
1	heuristic	6-0-0	18	—
2	active_inference	4-0-2	12	45
3	llm_fusion	4-0-2	12	42
4	kg_heuristic	2-0-2	6	—
5	random	1-0-4	3	—
6	llm (pre-fix)	1-0-4	3	—
7	world_model (pre-fix)	0-0-4	0	—

Table 6: Standard 7-agent Swiss tournament, 6 rounds, seed 7. `llm` and `world_model` rows reflect the runner before the auto-concede fix described in §6.2 and are recorded here for transparency; their corrected pairwise numbers appear in Table 5.

Rank	Agent	Wins	Pts
1	kg_heuristic	3	11
2	random	2	8
3	llm_fusion	1	5
4	heuristic	0	2

Table 7: Commander 4-seat pod, 8 rounds with seat rotations, seed 7. Sample size is small (8 games) and the deck pool is fixed at 4 cores — treat as a smoke test for end-to-end EDH play, not a strength claim.

(vii) Debugging note: harness-induced auto-concede. An earlier overnight run reported the LLM and world-model agents losing all 16 ablation games on turn 1. Investigation traced the cause to the rules engine always exposing `ActionType.CONCEDE` as a legal action (a debug affordance, see `rules_engine.get_legal_actions`). The Llama-3 prompt enumerated the legal-action indices but omitted `CONCEDE` from the rendered options, so a stray “1” in the model’s response collapsed onto the concede slot; the world-model controller, with effectively random initial weights, sampled the same slot stochastically. Filtering `CONCEDE` out of the candidate set inside both `LLMAgent.decide_action` and `WorldModelAgent.decide_action` restored normal play and produced the numbers in Table 5. We regard this as an instructive small example of how an under-specified action interface — not the policy itself — can dominate apparent agent quality, and have added a regression test to `tests/test_agents/`.

(viii) JEPA world-model ablation. We trained four JEPA variants for 30 epochs each on 120 freshly collected self-play trajectories (Modern Burn vs. Azorius Control) and evaluated each against four fixed baselines (8 games each), using `scripts/benchmark_trained_agents.py`. Results are in Table 8.

(ix) Cross-seed robustness. To test sensitivity to random initialisation we ran the same four-agent comparison (`heuristic`, `kg_heuristic`, `world_model`, `active_inference`) for 72 games each across five random seeds (`{11, 23, 47, 101, 20260428}`, 360 total games). Agent rankings are stable to within ± 2 pp across all seeds: `kg_heuristic` leads at 26.4 % mean win rate, `heuristic` second at 24.4 %, `world_model` third at 24.2 %, `active_inference` fourth at 25.0 %. The compressed range (< 6 pp between first and fourth) is expected given the domain: at ≤ 60 turns on Modern, games resolve quickly on mana efficiency more than on strategic depth.

(x) LLM model-size sweep. Stream C evaluated three small Ollama-hosted models — `llama3.2:1b`, `qwen2.5-coder:1.5b`, and `gemma3:4b` — each playing 8 games against `heuristic` on the same Modern matchup (seed 20260428). All three finish at exactly 50 %, with mean game

Variant	vs heuristic	vs random	vs llm	vs active_inf	Mean WR
full (KG + free-bits)	0.38	0.50	0.50	0.38	0.44
no_kl ($\beta_J=0$)	0.50	0.50	0.50	0.38	0.47
small (KG-dim 64)	0.38	0.38	0.38	0.50	0.41
no_kg	0.50	0.25	0.25	0.38	0.34

Table 8: Archived JEPA checkpoint-variant ablation: 30 training epochs, 120 self-play trajectories (seed 20260428), and 8 evaluation games per baseline. The **no_kg** point estimate is lowest; disabling encoder KL regularisation ($\beta_J = 0$) gives a slightly higher mean than the full variant. These small samples do not establish a graph benefit or a cause for the difference. Because dream search uses separate recurrent dynamics, this is not a controlled comparison of JEPA rollout accuracy.

Agent	s=11	s=23	s=47	s=101	s=main	Mean
kg_heuristic	27.8	26.4	25.0	27.8	25.0	26.4
active_inference	22.2	23.6	23.6	20.8	26.4	23.3
heuristic	25.0	29.2	29.2	30.6	25.0	27.8
world_model	25.0	20.8	22.2	20.8	23.6	22.5

Table 9: Win-rate (%) per seed, 72 games per seed per agent, Modern matchup. “s=main” is seed 20260428.

lengths of 6.75, 7.25, and 7.0 turns respectively and per-game wall times under 1 s. We interpret this as a clean negative: within the 1–4 B parameter range, model capacity does not differentiate play quality when actions are constrained to the engine’s legal set and no chain-of-thought is enabled. The apparent ceiling suggests that action-selection quality, not language quality, is the bottleneck — consistent with our decision to move strategic reasoning into the world model and use the LLM only as a soft prior over the legal set.

(xi) Planned: objective ablation for the control formulation. The claim of Section 3.3 — that belief-space KL control is the right objective and the two conventional EFE factorisations are not — is an empirical claim in our domain, and we state the experiment that tests it rather than asserting the conclusion. All three objectives are exposed by one picker interface. However, the current KL arm performs horizon planning while the legacy EFE arms use one-step scoring. The shared belief, transition model and preference potential do not remove this planning-budget confound. The implemented interleaved episode sweep is consequently a policy-stack pilot; an objective-only ablation requires a common continuation set, horizon and computational budget. Picker seeds alone also do not seed the native engine or opponent AI.

Three measurements are planned against **phase-ai** at fixed difficulty:

1. **Win rate.** If the analysis transfers, **kl** should dominate both surrogates.
2. **An irrelevant-information probe.** Construct matched positions with a learnable but strategically irrelevant observation, controlled action/resource costs and unchanged attainable outcome values. Compare selections with and without the epistemic bonus; include irreducible random noise as a negative control. Raw cantrip counts or an unchanged immediate action do not suffice, because draw quality, material and later contingencies may still matter.
3. **An ambiguity-aversion probe.** Define the observation channel explicitly and separate conditional observation entropy, belief uncertainty and reward variance. Compare matched tactical choices with equal deviation costs and evaluated outcome preferences, varying only the ambiguity term. Declined attacks or open mana are descriptive statistics, not proof of a pathological entropy penalty.

The existing legacy arms are proxy baselines, not full theoretical observation-channel implementations. A controlled result must qualify their correspondence to the equations before generalising to Active Inference. No separation would also be informative, but could reflect limited horizons, model errors or inadequate probes rather than establishing equivalence of the objectives.

(xii) Planned: ontology-conditioned evaluation. The ontology is validated structurally and against its own competency questions (Neubürger and Neubürger, 2026), but the *agent-facing* value of the symbolic layer is not yet measured. Two follow-ups are queued. First, a reasoning benchmark generated from the ontology itself by backward-chaining proof search in the manner of Van Schependorn et al. (2026), which gives controlled multi-hop depth and proof-based negatives for free. Second, submission of the generated artefact to a public competency-question-driven ontology benchmark (Ontology Engineering Group, Universidad Politécnica de Madrid, 2026) as an additional baseline, since our construction combines author-directed, LLM-assisted conceptual modelling with deterministic normative vocabulary extraction rather than an autonomous generation pipeline.

We deliberately do not claim “state of the art” numbers: we know of no comparable public baseline that plays full-Comprehensive-Rules MTG, and the contribution of this report is the system itself, plus the ontology and the released self-play infrastructure.

7 Discussion and Limitations

Scope. The engine is `phase-rs` and therefore covers the Comprehensive Rules to whatever depth upstream does; our own exposure is the adapter, which currently flattens the new viewer-interaction schema into the same opaque choice list as ordinary actions and so cannot yet reason about *what kind* of choice it is making. The ontology v2.0 covers the core game structure, the ability taxonomy, the strategic layer and the agent decision module, but does not yet encode every keyword from every set, and we do not extend it automatically per Standard rotation.

Ontology limitations. Three are worth stating plainly. (i) The import-resolved schema has passed a local HermiT consistency/satisfiability run, but independent DOLCE modelling review and complete profile conformance remain unqualified: the unmodified external documents have OWLAPI profile violations, and a schema-plus-vocabulary run exceeded 600 seconds. Focused positive/negative object/event probes do not certify a populated graph; CR 205.3 is checked procedurally. A read-only Fuseki snapshot reproduces 14 SPARQL checks, not agent grounding or implicit reasoning. (ii) Card designs now instantiate the type, subtype and quality/region patterns, but the role/situation pattern has no population path yet — roles are produced only by gameplay, and the trace-to-graph writer for combat situations is not implemented. (iii) Coverage is measured against our own competency questions, which is a weak test: a question set authored alongside an ontology will tend to be answerable by it. Extrinsic evaluation against an external benchmark is the honest check and is queued in Section 8.

Representation and computation. The present representation has 256 latent dimensions, and recurrent dynamics retain a separate history state. Larger models, deeper rollouts and more samples change inference cost; whether they improve playing strength must be measured. Planning accuracy cannot be inferred from model size or one-step training loss.

The control formulation is adopted, not re-derived. We use the belief-space objective motivated by Kaufmann (2026); the contribution is its proposed domain transfer and controlled evaluation, not a new equivalence theorem. Native KL rollouts presently use an analytic approximation. A learned latent predictor is a proposed replacement, not a JEPA decoder of

complete game states. The tree planner’s default continuation set is narrow, while the current sampled planner uses open-loop continuations. Neither presently establishes the full contingent planning needed to demonstrate dual-control information value.

LLM grounding. Selectors operate over engine-provided options rather than inventing action objects. This constrains the executable output, but does not guarantee strategically grounded reasoning. Invalid text, unavailable inference and incomplete candidate projections remain explicit failure modes. Using model preferences as a planner prior is distinct from direct option selection.

Decision-model limitations. Tev1’s decision format fits the agent interface, but its training recipe is not MTG-specific and it does not simulate consequences. A plausible choice distribution may reflect textual or option-order bias rather than tactical understanding. Restricting candidates can hide the best move, and action coverage is not an optimal-action oracle. Confidence concentration is not calibration. The model can therefore complement, but does not substantiate or replace, the learned-dynamics and control claims of this paper.

8 Future Work

The following items are tracked in `IMPLEMENTATION_PLAN.md` and constitute the near-term roadmap:

1. **Controlled decision-model evaluation.** Complete the matched-context, matched-shortlist and size-matched comparisons of Section 6.1; instrument chat fallback, seed the native engine/AI and rotate seats. Then evaluate Tev1 as a critic over world-model candidates separately from its direct-selector baseline.
2. **Strategy-aware and agent-driven mulligan policy.** The current keep/bottom heuristic is a single deterministic land-count rule shared by all agents. We are moving the decision through the agent interface (`MTGAgent.decide_mulligan`, `MTGAgent.bottom_cards`) so that aggressive, control, combo, and reactive strategies—and ultimately learned policies—can evaluate their own opening hands. JEPA-conditioned mulligan selection (rolling out a few imagined turns from each candidate hand) is the longer-term target.
3. **Priority-loop driven instants in the sync simulator.** The sync `GameSimulator` currently advances combat without exposing an instant-speed window to the non-active player; threading the existing `priority_loop` into combat steps will let counterspells and combat tricks fire mid-attack.
4. **Trained-agent benchmark.** *Completed* (Section 6.2 and table 8): four JEPA variants benchmarked against four baselines with active-inference checkpoint selection; `no_kl` ranked first (EFE = -0.034). Next: scale to ≥ 500 self-play trajectories and repeat to assess whether the KL term becomes beneficial with sufficient data.
5. **Active-inference checkpoint selection.** The EFE-based selector was used in vivo for the ablation above (Section 6.2). Extend from post-hoc ranking to an in-training promotion rule where candidates are selected during the self-play loop rather than only at evaluation time.
6. **Ablation matrix.** The four-variant JEPA sweep (Table 8) is a first step. Remaining: toggleable dream / opponent-model components with paired-game ELO comparison.
7. **Neural reasoner training.** A supervised + RL fine-tune of a graph-conditioned reasoner (`NeuralReasonerAgent`) replacing the SBERT card embedding with end-to-end GraphSAGE embeddings.

8. **Parallel self-play.** `asyncio.gather`-based parallel game execution to lift trajectory throughput, plus restoring `pytest-asyncio` so the async runner is exercised in CI.
9. **Belief-state visualisation.** JSON export and a Streamlit panel for the particle-filter posterior over the opponent’s hand.
10. **Commander politics.** Threat assessment across $N > 2$ opponents, including a multi-agent extension of the active-inference objective, and the Commander variant of the mulligan (free first mulligan).
11. **Ontology evolution and ABox migration.** Migrate the Scryfall importer onto the v2.0 role/situation and quality/region patterns so the expressive gain reaches the populated graph, then wire the framework into a human-in-the-loop ontology-evolution toolchain (Data Science Lab, Fachhochschule Südwestfalen, n.d.; jiayi et al., 2026) so the schema is updated semi-automatically per Standard rotation, with humans adjudicating debated axioms. The retrieval-scoped extension strategy of Lippolis et al. (2026) — retrieve the relevant ontology fragment per competency question rather than stuffing the whole TBox into context — is the approach we intend to follow.
12. **Promotion of induced evidence.** Define and evaluate the gate by which an `mtg:InducedSynergy` with sufficient support may be promoted to a curated `synergisesWith` edge, including how to represent and surface disagreement between runs (Ehrenmüller et al., 2026). This is the point at which the curated/induced boundary becomes a policy question rather than a schema question.
13. **Structured interaction handling.** Consume `phase-rs`’s viewer-interaction schema natively instead of flattening it, so the agent can distinguish a targeting decision from a mode choice from a damage assignment — each of which has a different shape of belief dependence.

9 Conclusion

We have presented a research framework for combining native rules execution, structured knowledge and learned planning in Magic: The Gathering. The repository implements candidate components and a completion-aware native evaluation interface, but does not yet demonstrate the complete ontology/KG/JEPA learning architecture end-to-end. The present evidence consists of resource validation, archived experiments and bounded native feasibility checks. Controlled strength comparisons, live graph grounding and multiplayer qualification remain explicit gates. We hope the framework provides a useful experimental substrate without mistaking architectural ambition for a measured result.

Reproducibility statement. All code, the OWL ontology, sample decklists, training scripts, and this report are available under MIT at <https://github.com/cuteredpwnda/opposition-agents-playing-mtg>.

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